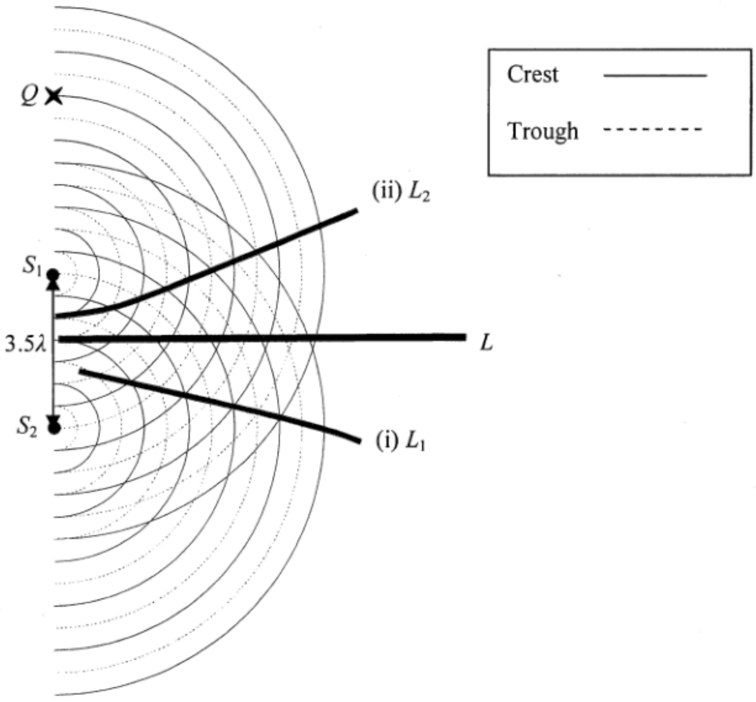
6. (a)

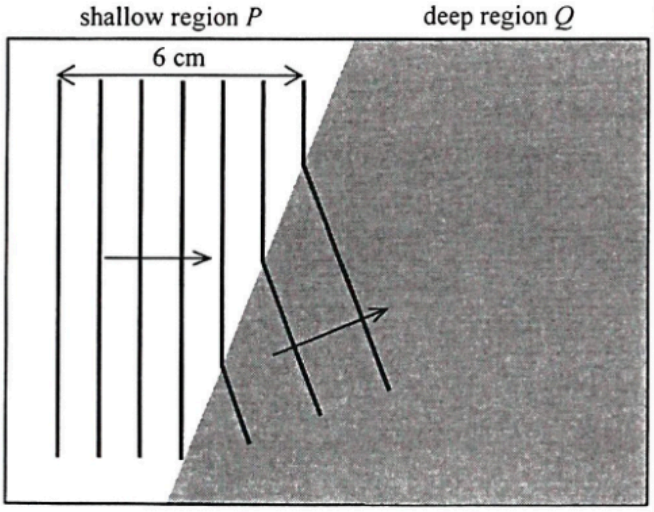


2A

L_1 / L_2 are further from L or separation of L /L_1 /L_2 is greater or angle between L /L_1 /L_2 larger. 1A <u>3</u>

7. (a) (i)	$\tan\theta = 0.38$ $\theta = 20.8°$	1A	Accept 20.8° to 21°
		1	
(ii)	$d\sin\theta = n\lambda$		
	As $d = (\frac{1}{300}\times10^{-3})$,	1M	
	$(\frac{1}{300}\times10^{-3})\times\sin 20.8° = 2\lambda$	1M	
	$\lambda = 5.92\times10^{-7}$ m (or 592 nm)	1A	Accept 5.90×10^{-7} m to 5.97×10^{-7} m
		3	
(iii)	Smaller percentage error in x / the diffraction angle θ.	1A	
	$\boxed{\text{Or Larger } x\text{, percentage error decreases.}}$	1A	
		1	
(b)	Repeat the procedures with the pin on the left-hand side / the other side of the observer.	1A	
	Take the average value of x obtained from both sides to calculate λ.	1A	
		2	

Solution	Marks	Remarks
7. (a) (i) Increase the separation between the double slit and the screen, D.	1A	
	1	
(ii) The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
	1	
(iii) The angular position of the 2nd order bright dot $\theta = \tan^{-1}\left(\frac{1.56/2}{1.40}\right) = 29.124053°$	1M	
Grating spacing $d = \frac{10^{-3}}{400} = 2.5 \times 10^{-6}$ m	1M	
Applying $d\sin\theta = n\lambda$, Wavelength $\lambda = \frac{2.5\times10^{-6}\times\sin 29.12°}{2}$ $= 6.08378 \times 10^{-7}$ m $\approx 6.08\times10^{-7}$ m (= 608 nm)	1A	Accept $\lambda = (6.06 \text{ to } 6.10)\times10^{-7}$ m
	3	
(b) (i) The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength << separation of the two sources), OR λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources << separation of the sources and detector), OR a is much smaller than D Any ONE	1A	Note : the order of magnitude of the wavelength of sound is about 10^{-1} m
Or Using the fringe separation equation to find the wavelength of sound is not accurate for - λ is comparable to/ NOT much smaller than a - a is NOT much smaller than D Any ONE	1A	
	1	
(ii) For the 1st order maximum, wavelength λ = path difference $PB - PA$	1M	
$= \sqrt{(1+0.5)^2+2^2} - \sqrt{(1-0.5)^2+2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438$ m	1M	
speed of sound: $v = f\lambda = 750 \times 0.4384$ $= 328.835 \text{ m s}^{-1}$ $\approx 329 \text{ m s}^{-1}$	1A	Accept $v = 328 \text{ m s}^{-1}$ to 330 m s^{-1}
	3	

5.	(a)	(i)	wavelength $\lambda = \dfrac{0.06}{7-1}$ $= 0.01$ m (= 1 cm)	1A	
				1	
		(ii)	speed $v = f\lambda = 10 \times 0.01$ $= 0.1$ m s^{-1} (= 10 cm s^{-1})	1M/1A	
				1	
	(b)	(i)	frequency = 10 Hz	1A	
				1	
		(ii)		1A 1A	
				2	
		(iii)	Refraction. It is due to the change in wavelengths / wave speeds in different media / depths.	1A 1A	
				2	

6.	(a)	Apply $d\sin\theta = m\lambda$, where θ is the deviation angle of the diffracted beam, m is the order of diffraction maximum. $\tan\theta = \frac{x/2}{L} \Rightarrow \theta = 24.702430°$ or $\sin\theta = \frac{x/2}{\sqrt{L^2 + (x/2)^2}} = 0.417906$	1M		
		$d = \frac{\lambda}{\sin\theta} = \frac{650\times10^{-9}}{0.418}$	1M		
		$= 1.555375\times10^{-6}$ m ≈ 1.56 μm	1A		Accept: 1.56 μm ~ 1.6 μm
				3	
	(b)	x will decrease as λ decreases, $\sin\theta$ and θ will decrease.	1A 1A		
				2	